

Experiment 'p_2_7_circle_properties' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_7_circle_properties

Natural language statement: Write a method that takes a radius and then returns the area and circumference of a circle with that radius and the volume and surface area of a sphere with that radius.

Method signature: p_2_7_circle_properties(radius: real) returns (circumference: real, area: real, volume: real, surface_area: real)

Ensures

- `circumference == 2.0 * 3.14159 * radius`
- `area == 3.14159 * radius * radius`
- `volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius`
- `surface_area == 4.0 * 3.14159 * radius * radius`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a radius and then returns the area and circumference of a
→ circle with that radius and the volume and surface area of a sphere with that
→ radius.

The signature should be:

p_2_7_circle_properties(radius: real) returns (circumference: real, area: real, volume:
→ real, surface_area: real)

The method should respect the following contract:

ensures circumference == 2.0 * 3.14159 * radius, ensures area == 3.14159 * radius *
→ radius, ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius,
→ ensures surface_area == 4.0 * 3.14159 * radius * radius

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXvR8NHgstmZSiH7DF20dhpOxkU

Model: gpt-5.5-2026-04-23

Created at: 1780484131

Finish reason: stop

Usage: 456 tokens in, and 324 tokens out

```
{
  circumference := 2.0 * 3.14159 * radius;
  area := 3.14159 * radius * radius;
  volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;
  surface_area := 4.0 * 3.14159 * radius * radius;
}
```

Final Program

```
method p_2_7_circle_properties(radius: real) returns (circumference: real, area: real,
→ volume: real, surface_area: real)
  ensures circumference == 2.0 * 3.14159 * radius
  ensures area == 3.14159 * radius * radius
  ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius
```

```
        ensures surface_area == 4.0 * 3.14159 * radius * radius
{
    circumference := 2.0 * 3.14159 * radius;
    area := 3.14159 * radius * radius;
    volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;
    surface_area := 4.0 * 3.14159 * radius * radius;
}
```

Total Token Usage

Input tokens: 456

Output tokens: 324

Reasoning tokens: 226

Sum of 'total tokens': 780

Experiment Timings

Overall Experiment started at 1780484131654, ended at 1780484142138, lasting 10484ms (10.48 seconds)

Iteration #1 started at 1780484131654, ended at 1780484142138, lasting 10484ms (10.48 seconds)